The Verification Gap: Checking the Reasoning in AI-Assisted Academic Writing

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paper1-draft-1-27-g482a33c-dirty

Abstract

Generative AI tools now help write a growing share of academic papers. Checking whether their citations exist has received attention, but a paper whose contribution is an argument can cite only real sources and still reason badly. Reporting guidelines appear to offer little help here. The EQUATOR (Enhancing the QUAlity and Transparency Of health Research) Network lists about 700 guidelines for empirical health research, and I found no consensus checklist for verifying the reasoning of non-empirical papers. This commentary argues that the vocabulary for such a checklist already exists. Toulmin's layout of arguments treats every assertion as a claim that rests on grounds through a warrant and carries a qualifier; Whetten's criteria judge a theoretical contribution as a whole. Combined, they allow a procedure that I propose: register each claim with what it rests on, check each item by its kind, and check the contribution once. The paper applies this procedure to itself and publishes the record. It reports no evaluation; whether the procedure helps authors and reviewers is an open question.

Keywords: academic writing, generative AI, verification, argumentation, reporting guidelines, non-empirical research

Key Points:

- Generative AI tools can produce fluent papers whose citations are real and whose reasoning is still weak.
- Reporting guidelines cover empirical research; I found no consensus checklist for verifying the reasoning of non-empirical papers.
- Toulmin and Whetten supply a vocabulary for such a checklist: each claim is checked by what it rests on, and the contribution is checked once.
- The paper publishes its own claim registry as a worked example; it does not show that the procedure works.

1 Introduction

Writing assistants based on large language models (LLMs) have spread quickly: up to 22% of computer science papers now show signs of LLM editing, and such editing can be detected in every major discipline (Liang et al., 2025). In this paper "generative AI tools", or "AI" for short, means these assistants. They write fluently, and they can synthesise literature, structure an argument, draft prose and format references. When a human author falls short, the text tends to show it, in awkward phrasing, gaps in coverage or inconsistent formatting. An AI-assisted draft can read as polished and complete while it contains errors that only checking would reveal, such as a citation to a paper that does not exist.

Citation hallucination is the most discussed of these errors. Mugaanyi et al. (2024) found that 62–89% of the digital object identifiers (DOIs) that LLMs generated for natural-science and humanities topics were fabricated, and Walters and Wilder (2023) found fabricated citations in 55% of references generated by OpenAI's GPT-3.5 model and 18% of those by GPT-4 in generated literature reviews. A fabricated citation cannot be caught by knowing the field, because the work it cites does not exist; it can be caught by looking the work up.

A second error is harder to see. I call it *confidence inflation*: wording that expresses more certainty than the evidence supports. The evidence indicates that LLMs verbalise high confidence even on questions they answer wrongly (Xiong et al., 2024; Zhou et al., 2024) and that their summaries of scientific work generalise findings beyond what the original studies claim (Peters and Chin-Yee, 2025). Whether this carries over into academic prose is unsettled: in one corpus of applied-linguistics abstracts, ChatGPT used boosters such as "shows" at twice the human rate, but also more hedges (El-Dakhs et al., 2025). The stakes are epistemic, because words such as "demonstrates" and "suggests" are meant to tell a reader how much weight a claim can bear.

In technical papers a third error warrants investigation: derived numbers that look plausible and are wrong. In informal trials on my own projects, prompting a model to recompute every value caught arithmetic errors that prompting it to judge soundness

missed; this is an observation, not a measured result.

Recent systems go beyond checking that references exist. A linter checks a manuscript's references, internal consistency and causal language against the study design (Samsonau, 2026); a workflow pairs each drafted claim with an evidence packet and returns unsupported claims to the author (Chen et al., 2026); a provenance discipline binds reported quantities to an audit trail (Zhou and Yu, 2026); and a checklist for verifying cited assertions in narrative reviews is in development (VANRA Development Group, 2026). Each asks whether evidence exists, supports a claim or suffices, and the linter also checks causal wording against the study design. None that I found asks whether a warrant holds for its readers, or chooses the check by what a claim rests on, which is where a paper whose contribution is an argument can fail.

Detecting AI-written text does not fill that gap. Detection asks who wrote a text; verification asks whether it holds. Detectors have also proved fragile in tests: their accuracy varies with generation settings (Dugan et al., 2024), and paraphrase can collapse it (Krishna et al., 2023).

2 Reporting Guidelines and Non-Empirical Papers

A well-developed example of verification infrastructure for academic writing is the EQUATOR Network, which lists about 700 reporting guidelines for health research (EQUATOR Network, n.d.). Guidelines such as CONSORT for randomised trials and PRISMA for systematic reviews (Page et al., 2021) give authors a checklist to complete while they prepare a manuscript, and the GoodReports tool picks the checklist that fits a study type (Struthers et al., 2021). For one guideline the effect has been measured: journals endorsing CONSORT reported more completely on 25 of 27 checklist items, significantly so for five (Turner et al., 2012), and incomplete reporting is a recognised source of research waste (Glasziou et al., 2014).

Guidance for papers without new data exists, but it is scattered and seems not to reach the reasoning. Conceptual papers lack "commonly accepted templates" (Jaakkola, 2020, p. 18), although Jaakkola proposes four. Design-science researchers have reported

no agreed guidance on presenting their work (Gregor and Hevner, 2013; Dinter and Krawatzeck, 2015), and validity in that field "is neither well-understood nor consistently established" (Larsen et al., 2025). Software engineering's community standards ask methodological papers for "clear, valid arguments" but give no procedure for checking them (Ralph et al., 2020). A quality scale for narrative reviews exists, though its authors call it "not a reporting guideline" (Baethge et al., 2019), and reporting items for concept analysis have only recently been collected (Nashwan et al., 2026). I found no consensus checklist for verifying the reasoning of such papers: whether each step from evidence to claim holds.

Gregor (2006) distinguishes five types of theory in information systems research. Two of them, analytic theories (Type I) and theories for design and action (Type V), contribute conceptual structures more than findings, and such contributions may not be checkable against an earlier source, because they are the new contribution. Theoretical, conceptual, design-science and perspective papers are examples of non-empirical papers in this sense; the list is not a classification.

Building on Toulmin (2003) and Whetten (1989), I suggest that for these papers the main question may be less "do the data support the claim?" and more "does the reasoning hold?". That would be a different task from the one reporting guidelines support.

3 Toulmin and Whetten as a Basis for Verification

Two well-known frameworks may offer much of what a checklist for reasoning needs. Box 1 gives the terms; the framework's glossary defines them from their sources.[1]

Box 1. Terms

Claim: any assertion put forward for acceptance. **Grounds**: what a claim rests on (Toulmin's *data*). **Warrant**: the step from grounds to claim. **Qualifier**: the hedge that states how strong the claim is. **Rebuttal**: circumstances in which the step no longer holds. **Declared starting point**: an assumption, definition or value the paper adopts without arguing for it. **Contribution**: what the paper adds; checked once, as a whole. **Verification**: checking each claim against its grounds and warrant. **Validation**: checking that the contribution serves its readers and purpose.

[1]The glossary, the claim registry and the checking tools are public at `https://github.com/ducroq/agent-ready-papers`.

Toulmin modelled his layout of arguments on jurisprudence. He called logic "generalised jurisprudence" and likened the difference between data and warrant to "the distinction drawn in the law-courts between questions of fact and questions of law" (Toulmin, 2003, pp. 7, 92). In his account any assertion is a claim; its *data* are "the facts we appeal to as a foundation for the claim"; its warrant is one of the "general, hypothetical statements, which can act as bridges" from data to claim; a *qualifier* indicates "the strength conferred by the warrant on this step"; and a *rebuttal* names "circumstances in which the general authority of the warrant would have to be set aside" (Toulmin, 2003, pp. 90–94). Toulmin later wrote that he had not had in mind an analytical model like the one that came to be called "the Toulmin model"; his concern "was with twentieth-century epistemology, not informal logic" (Toulmin, 2003, p. vii). Using his layout as a checklist is my adaptation, and it has to be judged on its own merits.

That adaptation may be workable. LLMs can extract Toulmin's elements from text when prompted with them (Gupta et al., 2024), which is extraction and not yet verification, so a tool may help an author lay out what each claim rests on.

The procedure I propose treats a paper's claims as differing less in kind than in what they rest on. The grounds of a claim may be a cited source, other claims in the paper, a declared starting point such as a definition or a value, or the author's own observation. In the procedure, each kind of grounds gets its own check:

- a **source** is checked in two steps: the work exists, and it says what the claim attributes to it;
- a step from **other claims** is checked by asking whether the warrant holds for the paper's readers;
- a **declared starting point** is checked only for being stated, with a note on whether the field accepts or contests it;
- an **own observation** is checked for a method someone else could repeat.

Combining several sources into a point none of them makes alone adds an inference, which needs the warrant check. A recommendation needs two things more: a declared value it

serves, and the conditions under which it applies.

Whetten's criteria work at a different level. He set out what "a complete theory must contain": which factors (What), how they relate (How), why ("the theoretical glue that welds the model together"), and for whom, where and when it holds (Whetten, 1989, pp. 490–492). His seven reviewer questions, from "What's new?" to "Who cares?", answer "What constitutes a publishable theory paper?" (Whetten, 1989, p. 494). They judge a contribution as a whole, so they may serve best as a check of the paper's central contribution, applied once, and not of each statement in it.

These two levels map onto a distinction from systems engineering. Verification shows "proof of compliance with requirements"; validation shows that a product "accomplishes the intended purpose in the intended environment" (NASA, 2016, §2.0). Checking each claim against its grounds and warrant is verification in this sense; asking Whetten's questions of the contribution is closer to validation. The mapping is mine.

Qualifiers need a scale. Fixed wording for levels of certainty has precedents. Kent (1964) argued for "a few key odds expressions" in intelligence estimates; the Intergovernmental Panel on Climate Change (IPCC) asks its authors to judge confidence from the "type, amount, quality, and consistency of evidence" and "the degree of agreement" (Mastrandrea et al., 2010); and the GRADE working group (Grading of Recommendations Assessment, Development and Evaluation) fixes wording per level of certainty and size of effect, from "results in" to "may result in" (Santesso et al., 2020). Table 1 adapts this practice with four confidence tiers, each permitting certain qualifiers. It assigns tiers mainly by counting agreeing sources, which is simpler than the IPCC's two axes and has not been tested with readers, as GRADE's wording was. It may still help an author notice when a sentence claims more than its grounds allow.

Table 1: Confidence tiers and the qualifiers they permit.

Tier	**Assigned when. . .**	**Qualifiers**
Established	Multiple independent sources; no dispute	"shows", "confirms"
Supported	Two or three sources agree; open questions remain	"indicates", "evidence suggests"
Emerging	One or two sources; not replicated	"may", "preliminary evidence"
Speculative	Inference only; no direct evidence	"warrants investigation"

Every claim also carries a *priority*, the author's answer to the question of what fails if the claim is wrong. A P0 claim is one the paper fails without; P1 strengthens the argument; P2 gives context. The priority sets how complete the checking must be, and every P0 claim is expected to reach the Supported tier.

The central argument of this paper builds on these pieces. If a claim is checked by a procedure that does not fit what it rests on, the check may fail it for the wrong reason: an inference judged by whether it has a source looks unsupported even when its reasoning holds. Recording each claim's grounds item by item, and letting the kind of grounds decide the check, may avoid such false failures. The argument rests on design reasoning and on this paper's own registry; it has not been tested across authors or fields.

4 Limitations

The vocabulary leaves what is probably the hardest part, judging whether a warrant holds, to the reader. Research on argument quality offers more structure. Wachsmuth et al. (2017) distinguish logical, rhetorical and dialectical quality and define a cogent argument as one with "individually acceptable premises that are relevant to the argument's conclusion and, together, sufficient to draw the conclusion"; Walton et al. (2008) collect argumentation schemes, which their publisher describes as "forms of argument that capture stereotypical patterns of human reasoning". Either could make the warrant check more specific than Toulmin's layout does.

Whether tools can carry part of the load is also open. Structured prompting was reported to improve LLM argument extraction (Gupta et al., 2024), and representing support and attack formally to improve LLM review aggregation (Sukpanichnant et al., 2024). Both are narrow tasks, so they suggest only that structure combined with LLMs may outperform LLMs alone.

The tier rule is a simplification. Counting sources does not measure their independence or quality, and it gives no guidance for an inference, whose strength lies in its warrant. And the procedure has one user so far: the author of this paper. This commentary is a first step that proposes a way of checking. It does not show that the way works, which

needs authors and reviewers in several fields to use it and a comparison with papers checked without it.

5 Recommendations

I propose that the scholarly community develop verification infrastructure for non-empirical papers, as EQUATOR did for empirical health research. The value behind the proposal is that published reasoning should be checkable: readers and reviewers should be able to see what each claim rests on. Its grounds are the gap described in Section 2 and the argument of Section 3. A first version could be modest: authors register the claims a paper depends on, with their grounds, warrants and qualifiers, and publish the register with the paper, as this one does.

The proposal applies to academic papers whose contribution is an argument, and to the interpretive parts of empirical papers. The same procedure might serve other writing that asks readers to trust its reasoning, journalism among them, but that is untested. Toulmin's and Whetten's frameworks come from Western academic traditions; their fit with other scholarly conventions needs to be assessed.

Peer review already judges arguments, and a register would not replace it. It could help authors find weak steps before submission and help reviewers see where a paper's weight rests.

Several directions warrant investigation: testing whether registers change what reviewers find, adapting the checks to the conventions of other fields, building them into writing tools so that the checks run while an author drafts, and recognising the recomputation of derived values as a check of its own.

Acknowledgements and use of AI

[**Author to write: roles.**] Who originated the ideas, the structure and the judgements, and what AI tools did (drafting, rephrasing, source search, checking). [**Author to write: tools.**] The models and services used, and for which tasks. Four models from different

vendors reviewed the first draft and the framework's glossary; their reviews informed this version. The author takes full responsibility for the content. [**Verification record, to be filled from the tool runs of the submitted version, with their date.**]

A Self-Application

This paper was written with the procedure it proposes. Its claim registry, published with the paper,[2] records each claim's grounds, warrant, qualifier, confidence tier and priority, plus the paper's declared starting points and its one contribution. Three entries show how it works.

A claim resting on sources (S1-1). "AI citation hallucination is a distinct failure mode." Grounds: two sources, Mugaanyi et al. (2024) and Walters and Wilder (2023), each checked in both steps. Priority P0: the paper's opening argument fails without it. Tier: Supported, because two peer-reviewed sources agree; the qualifier in the text is a reported finding ("found"), not "shows".

A claim resting on an inference (S2-4). "For non-empirical papers, argument quality ... is the primary verification challenge, not factual accuracy." Grounds: Toulmin (2003) and Whetten (1989). Neither says this; the step from what they say to the claim is mine, and the registry records it as a warrant. Priority P1: it strengthens the argument for a reasoning check but the paper does not fail without it. Tier: Emerging, so the text says "may". Under the earlier version of the framework this entry was typed as a sourced claim, and the inference was invisible.

The contribution (C1). Table 2 applies Whetten's elements. The contribution rests on three claims, S2-1 (Supported), S2-2 (Emerging) and S3-4 (Emerging), on one declared value (D1) and on three declared definitions (D2–D4), so its tier is Emerging.

[2] `https://github.com/ducroq/agent-ready-papers/blob/master/papers/perspective/vv/claims/claim_registry.md`

Table 2: Whetten's elements applied to the contribution (C1).

Element	Content
What	Verification infrastructure; non-empirical papers; AI-assisted writing (each a declared definition).
How	Non-empirical papers need such infrastructure to stay checkable as AI tools make fluent prose cheap.
Why	Guidelines exist for empirical research (S2-1), none found checks reasoning (S2-2), and each claim needs the checks its grounds call for (S3-4).
Who, where, when	Academic papers whose contribution is an argument, and the interpretive parts of empirical papers; other writing, such as journalism, untested; frameworks from Western traditions.
Value	Published reasoning should be checkable (declared starting point D1).
Objections	Peer review already judges arguments: a register would support review, not replace it.

What the registry shows about this paper. Of the six P0 entries, three reach the Supported tier or above and three do not: the gap claim (S2-2), the central argument (S3-4) and the contribution (C1), which rests on both. By the framework's own rule the paper does not pass its gate for critical claims. That is reported here because a framework that asks authors to state the strength of their evidence should state its own.

References

Baethge, C., Goldbeck-Wood, S., and Mertens, S. (2019). SANRA—a scale for the quality assessment of narrative review articles. *Research Integrity and Peer Review*, 4:5.

Chen, G., Yu, Y., and Wang, W. (2026). Evidence-ledger adjudication for claim-evidence traceability.

Dinter, B. and Krawatzeck, R. (2015). Towards a configurative publication schema for design science research. In *Proceedings of the 36th International Conference on Information Systems (ICIS 2015)*. Research-in-Progress.

Dugan, L., Hwang, A., Trhlík, F., Zhu, A., Ludan, J. M., Xu, H., Ippolito, D., and Callison-Burch, C. (2024). RAID: A shared benchmark for robust evaluation of

machine-generated text detectors. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12463–12492.

El-Dakhs, D. A. S., Afzaal, M., and Siyanova-Chanturia, A. (2025). A genre-based comparison of Chat-GPT-generated abstracts versus human-authored abstracts: Focus on applied linguistics research articles. *Corpus Pragmatics*, 10(1):1.

EQUATOR Network (n.d.). Enhancing the QUAlity and Transparency of health research. `https://www.equator-network.org/`. Accessed: 2026-03-03.

Glasziou, P., Altman, D. G., Bossuyt, P., Boutron, I., Clarke, M., Julious, S., Michie, S., Moher, D., and Wager, E. (2014). Reducing waste from incomplete or unusable reports of biomedical research. *The Lancet*, 383(9913):267–276.

Gregor, S. (2006). The nature of theory in information systems. *MIS Quarterly*, 30(3):611–642.

Gregor, S. and Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly*, 37(2):337–355.

Gupta, A., Zuckerman, E., and O'Connor, B. (2024). Harnessing Toulmin's theory for zero-shot argument explication. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024)*, pages 10259–10276.

Jaakkola, E. (2020). Designing conceptual articles: four approaches. *AMS Review*, 10(1–2):18–26.

Kent, S. (1964). Words of estimative probability. *Studies in Intelligence*, 8(4).

Krishna, K., Song, Y., Karpinska, M., Wieting, J., and Iyyer, M. (2023). Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*.

Larsen, K. R., Lukyanenko, R., Mueller, R. M., Storey, V. C., Parsons, J., VanderMeer, D., and Hovorka, D. S. (2025). Validity in design science. *MIS Quarterly*, 49(4):1267–1294.

Liang, W., Zhang, Y., Wu, Z., Lepp, H., Ji, W., Zhao, X., Cao, H., Liu, S., He, S., Huang, Z., Yang, D., Potts, C., Manning, C. D., and Zou, J. Y. (2025). Quantifying large language model usage in scientific papers. *Nature Human Behaviour*, 9(12):2599–2609.

Mastrandrea, M. D., Field, C. B., Stocker, T. F., Edenhofer, O., Ebi, K. L., Frame, D. J., Held, H., Kriegler, E., Mach, K. J., Matschoss, P. R., Plattner, G.-K., Yohe, G. W., and Zwiers, F. W. (2010). Guidance note for lead authors of the IPCC fifth assessment report on consistent treatment of uncertainties. Technical report, Intergovernmental Panel on Climate Change (IPCC).

Mugaanyi, J., Cai, L., Cheng, S., Lu, C., and Huang, J. (2024). Evaluation of large language model performance and reliability for citations and references in scholarly writing: Cross-disciplinary study. *Journal of Medical Internet Research*, 26:e52935.

NASA (2016). *NASA Systems Engineering Handbook*. National Aeronautics and Space Administration.

Nashwan, A. J., Kunjavara, J., Joy, G., Singh, K., Rony, M. K. K., Mannethodi, K., and Ahmed, S. K. (2026). Preferred reporting items for concept analysis in nursing: a systematic review. *Journal of Research in Nursing*, 31(5–6):498–511.

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., McGuinness, L. A., Stewart, L. A., Thomas, J., Tricco, A. C., Welch, V. A., Whiting, P., and Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372:n71.

Peters, U. and Chin-Yee, B. (2025). Generalization bias in large language model summarization of scientific research. *Royal Society Open Science*, 12(4):241776.

Ralph, P. et al. (2020). Empirical standards for software engineering research.

Samsonau, S. V. (2026). sciwrite-lint: Verification infrastructure for the age of science vibe-writing.

Santesso, N., Glenton, C., Dahm, P., Garner, P., Akl, E. A., Alper, B., Brignardello-Petersen, R., Carrasco-Labra, A., De Beer, H., Hultcrantz, M., Kuijpers, T., Meerpohl, J., et al. (2020). GRADE guidelines 26: informative statements to communicate the findings of systematic reviews of interventions. *Journal of Clinical Epidemiology*, 119:126–135.

Struthers, C., Harwood, J., de Beyer, J. A., Dhiman, P., Logullo, P., and Schlüssel, M. (2021). GoodReports: Developing a website to help health researchers find and use reporting guidelines. *BMC Medical Research Methodology*, 21:217.

Sukpanichnant, P., Rapberger, A., and Toni, F. (2024). PeerArg: Argumentative peer review with LLMs. In *Proceedings of NeLaMKRR@KR 2024*. arXiv:2409.16813.

Toulmin, S. E. (2003). *The Uses of Argument*. Cambridge University Press, updated edition. Originally published 1958.

Turner, L., Shamseer, L., Altman, D. G., Weeks, L., Peters, J., Kober, T., Dias, S., Schulz, K. F., Plint, A. C., and Moher, D. (2012). Consolidated standards of reporting trials (CONSORT) and the completeness of reporting of randomised controlled trials (RCTs) published in medical journals. *Cochrane Database of Systematic Reviews*, 2012(11):MR000030.

VANRA Development Group (2026). VANRA (verification of assertions in narrative review articles): a reporting checklist for narrative review articles. Reporting guidelines under development for systematic reviews, registered 2 September 2026. Contact: D. J. Parker. Guideline under development. Accessed 2026-09-26.

Wachsmuth, H., Naderi, N., Hou, Y., Bilu, Y., Prabhakaran, V., Alberdingk Thijm, T., Hirst, G., and Stein, B. (2017). Computational argumentation quality assessment in natural language. In *Proceedings of the 15th Conference of the European Chapter of*

the Association for Computational Linguistics: Volume 1, Long Papers, pages 176–187. Association for Computational Linguistics.

Walters, W. H. and Wilder, E. I. (2023). Fabrication and errors in the bibliographic citations generated by ChatGPT. *Scientific Reports*, 13:14045.

Walton, D., Reed, C., and Macagno, F. (2008). *Argumentation Schemes*. Cambridge University Press.

Whetten, D. A. (1989). What constitutes a theoretical contribution? *Academy of Management Review*, 14(4):490–495.

Xiong, M., Hu, Z., Lu, X., Li, Y., Fu, J., He, J., and Hooi, B. (2024). Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *The Twelfth International Conference on Learning Representations (ICLR)*.

Zhou, K., Hwang, J., Ren, X., and Sap, M. (2024). Relying on the unreliable: The impact of language models' reluctance to express uncertainty. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3623–3643.

Zhou, Y. and Yu, C. (2026). Auditable AI-assisted research writing: An engineering discipline with pre-registered process observation.